

Whole Exome Sequencing Analysis & Whole Mitochondrial Genome Sequencing

Patient name: XX	PIN: XX
Age: 3 Years	Gender: Male
Hospital/Clinic: XX	Sample Number: XX
Referring Clinician: XX	Sample Collection Date: NA
Specimen: Peripheral Blood	Sample Received Date: NA
	Report Date: NA

Clinical History

Proband, XX, was the first-born child to non-consanguineous parents. XX presented with bilateral temporal lens subluxation, high myopia, high-arched palate, arachnodactyly, positive wrist and thumb signs, dolichocephaly, and an arm span greater than height. Anthropometric evaluation revealed a weight of 13.5 kg and height of 108 cm. Cardiac evaluation showed an aortic root diameter at the sinus measuring 19 mm with a Z score of 1.5. Proband, XX was suspected to have Marfan syndrome and has been evaluated for pathogenic variations.

Whole Exome Sequencing Analysis

Results

Variant of uncertain significance causative to the related phenotype was identified

List of uncertain significant variant identified related to phenotype:

Gene	Variant*	Allele Status	Region	Disease or Disorder	Inheritance	Classification*
<i>FBN1</i> (-) (NM_000138.5)	c.5684G>T (p.Cys1895Phe)	Heterozygous	Exon 47	Marfan syndrome (OMIM# 154700)	Autosomal Dominant	Uncertain significance (PM2_Supp, PM5, PP3)

Variant Interpretation:

Gene	Variant*	Inheritance	Disease or Disorder	Criteria	Classification*
<i>FBN1</i> (-)	c.5684G>T (p.Cys1895Phe)	Autosomal Dominant	Marfan syndrome (OMIM# 154700)	PM2_Supp, PM5,PP3	Uncertain significance
Region	Allele Status	Sequence Ontology	Variant Frequency	Allelic Read Depths	
Exon 47	Heterozygous	Missense Variant	Not observed in annotated gnomAD v4 All	Ref(C): 106, Alt(A): 101, VAF: 48.79%	
In silico Prediction Tools				Conservation Status	
SIFT:Da	LRT:D	PolyPhen2:Da	CADD:D	MT2:D	PhyloP:C GERP++:C
Genomic Positor			#Clinical & Literature Evidences		
Chr15:NC_000015.10:g.48446810C>A (NM_000138.5)			ClinVar: 2925543 PubMed: 16222657		

Abbreviation: **D:** Deleterious; **Da:** Damaging; **Ds:** Disrupted; **C:** Conserved; **T:** Tolerated; **U:** Uncertain; **Supp:** Supporting;

#Clinical & Literature Evidences: A Missense Variant in same acid position (c.5684G>A, p.Cys1895Tyr) as pathogenic in ClinVar database. A missense variant in the same amino acid position (c.5683T>C, p.Cys1895Arg) has been previously reported in a patient affected with Marfan Syndrome and Other Fibrillinopathies in heterozygous state.

OMIM Phenotype: Marfan syndrome (OMIM#[154700](#)) is caused by heterozygous mutation in the *FBN1* gene (OMIM*134797). A heritable disorder of fibrous connective tissue, Marfan syndrome (MFS) shows striking pleiotropism and clinical variability. The cardinal features occur in 3 systems--skeletal, ocular, and cardiovascular. It shares overlapping features with congenital contractural arachnodactyly, which is caused by mutation in the *FBN2* gene. This disease follows autosomal dominant pattern of inheritance [2].

*Genetic test results are based on the recommendation of American college of Medical Genetics [1].

No other variant that warrants to be reported for the given clinical indication was identified

Whole Mitochondrial Genome Sequencing

Results

No pathogenic or likely pathogenic variant(s) causative of the reported phenotype was detected

*Genetic test results are based on the recommendation of American college of Medical Genetics [1].

No other variant that warrants to be reported for the given clinical indication was identified

CNV Findings

No significant Copy Number Variations (CNV) related to phenotype was detected.

Recommendations

- Sequencing the variant(s) in the parents and the other affected and unaffected members of the family is recommended to confirm the significance.
- Alternative test is strongly recommended to rule out the deletion/duplication.
- Genetic counselling is advised.

Methodology

DNA extracted from the blood was used to perform whole exome and whole mitochondrial using Exome + Mitochondrial capture kit. The targeted libraries were sequenced to a targeted depth of 80X to 100X using SURFSeq 5000 sequencing platform. This kit has deep exonic coverage of all the coding regions including the difficult to cover regions. The sequences

obtained are aligned to human reference genome (GRCh38.p13) using Sentieon aligner and analyzed using Sentieon for removing duplicates, recalibration and re-alignment of indels. Sentieon DNAscope has been used to call the variants. Detected variants were annotated and filtered using the VarSeq software with the workflow implementing the ACMG guidelines for variant classification. The variants were annotated using 1000 genomes (V2), gnomAD (v4.1.0,3.1.2,2.1.1), ClinVar, OMIM, dbSNP, NCBI RefSeq Genes. In-silico predictions of the variant was carried out using VS- SIFT, VS- PolyPhen2, PhyloP, GERP++, GeneSplicer, MaxEntScan, NNSplice, PWM Splice Predictor. Only non-synonymous and splice site variants found in the coding regions were used for clinical interpretation. Silent variations that do not result in any change in amino acid in the coding region are not reported. Gene list used in the assay.

The sequences obtained were aligned to revised Cambridge mitochondrial reference genome (rCRS) analyzed using sentieon to identify variants relevant to the clinical indication. We follow the Varseq best practices framework for identification of variants in the sample. Clinically relevant mutations were annotated using published variants in literature and Mito Map Database [Click here](#).

Sequence data attributes

Total Read Generated	13.54 Gb
Data ≥ Q30	96.12%

Genetic test results are reported based on the recommendations of American College of Medical Genetics [1], as described below:

Classification	Interpretation
Pathogenic	A disease-causing variation in a gene which can explain the patients' symptoms has been detected. This usually means that a suspected disorder for which testing had been requested has been confirmed
Likely Pathogenic	A variant which is very likely to contribute to the development of disease however, the scientific evidence is currently insufficient to prove this conclusively. Additional evidence is expected to confirm this assertion of pathogenicity.
Variant of Uncertain Significance	A variant has been detected, but it is difficult to classify it as either pathogenic (disease causing) or benign (non-disease causing) based on current available scientific evidence. Further testing of the patient or family members as recommended by your clinician may be needed. It is probable that their significance can be assessed only with time, subject to availability of scientific evidence.

Disclaimer

- The classification of variants of unknown significance can change over time. Anderson Diagnostics and Labs cannot be held responsible for it.
- Intronic variants, UTR and Promoter region variants are not assessed using this assay.
- This assay has a sensitivity of 70-75% in detecting large deletions/duplications of more than 10 base pairs or copy number variations (CNV). However, it is important to note that any CNVs detected must be confirmed using an alternate method.
- Certain genes may not be covered completely, and few mutations could be missed. Variants not detected by this assay may impact the phenotype.
- The variations have not been validated by Sanger sequencing.
- The above findings and result interpretation was done based on the clinical indication provided at the time of reporting.
- It is also possible that a pathogenic variant is present in a gene that was not selected for analysis and/or interpretation in cases where insufficient phenotypic information is available.
- Genes with pseudogenes, paralog genes and genes with low complexity may have decreased sensitivity and specificity of variant detection and interpretation due to inability of the data and analysis tools to unambiguously determine the origin of the sequence data in such regions. The variants of uncertain significance and variations with high minor allele frequencies which are likely to

be benign will be given upon request.

- In addition, this test does not cover detection of such structural variants in the genes *SMN1*, *SMN2*, *HBA1*, *HBA2*, *PMS2*, *GBA*, and *CYP21A2* which have very high homology. Moreover, the ORF15 transcript isoform (RefSeq id: NM_001034853.1) of the *RPGR* gene associated with retinitis pigmentosa is not sufficiently covered in the test.
- Incidental or secondary findings that meet the ACMG guidelines can be given upon request [3].

Reference

1. Richards, S, et al. Standards and Guidelines for the Interpretation of Sequence Variants: A Joint Consensus Recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. Genetics in medicine: official journal of the American College of Medical Genetics. 17.5 (2015): 405-424.

2. Amberger J, Bocchini CA, Scott AF, Hamosh A. McKusick's Online Mendelian Inheritance in Man (OMIM). Nucleic Acids Res. 2009 Jan;37(Database issue):D793-6. doi: 10.1093/nar/gkn665. Epub 2008 Oct 8.

3. Lee, Kristy et al. "ACMG SF v3.3 list for reporting of secondary findings in clinical exome and genome sequencing: A policy statement of the American College of Medical Genetics and Genomics (ACMG)." Genetics in medicine: official journal of the American College of Medical Genetics vol. 27,8 (2025): 101454. doi:10.1016/j.gim.2025.101454.

This report has been reviewed and approved by:

			
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Appendix 1: Gene Coverage Indication Based Analysis:

Gene Name	Percentage of coding region covered	Gene Name	Percentage of coding region covered	Gene Name	Percentage of coding region covered	Gene Name	Percentage of coding region covered
ABCC6	100.0	ABCC9	100.0	ADA2	100.0	ADAMTS17	100.0
ADAMTS18	100.0	ADK	100.0	AKT1	100.0	AKT2	100.0
AKT3	100.0	ALPL	100.0	ALX1	100.0	ALX4	100.0
ANKH	100.0	ANTXR1	100.0	ARL2	100.0	ARR3	100.0
ASPA	100.0	ASXL1	100.0	ASXL2	100.0	ATOH7	100.0
B3GALT6	100.0	BMP4	100.0	BRSK2	100.0	BRWD3	100.0
CCBE1	100.0	CCND2	100.0	CD96	100.0	CDC45	100.0
CDKN1C	100.0	CEP120	100.0	CHD3	100.0	CHD4	100.0
CHD8	100.0	CHST14	100.0	CHST3	100.0	CNGA3	100.0
COL11A1	100.0	COL11A2	100.0	COL2A1	100.0	COL3A1	100.0
COL4A1	100.0	COL5A1	100.0	COL9A1	100.0	COL9A2	100.0
COL9A3	100.0	COLEC11	100.0	CPSF1	100.0	CRB1	100.0
CTNNB1	100.0	CTSA	100.0	CTSH	100.0	CUL4B	100.0
CYP26B1	100.0	CYP4V2	100.0	D2HGDH	100.0	DHCR24	100.0
DICER1	100.0	DIS3L2	100.0	DNMT3A	100.0	DVL1	100.0
DZIP1	100.0	EED	100.0	EFNB1	100.0	EIF2B5	100.0
ELN	100.0	ENG	100.0	EPHA2	100.0	EPHB2	100.0
ERBB3	100.0	ERF	100.0	ESCO2	100.0	ETFA	100.0
ETFB	100.0	ETFDH	100.0	EXT2	100.0	EZH2	100.0
FAM20C	100.0	FBLN5	100.0	FBN1	100.0	FBN2	100.0
FGF3	100.0	FGFR1	100.0	FGFR2	100.0	FGFR3	100.0
FLT4	100.0	FLVCR2	100.0	FOXP1	100.0	FREM1	100.0
GATAD2B	100.0	GCDH	100.0	GFAP	100.0	GJA1	100.0
GJA8	100.0	GLI3	100.0	GLMN	100.0	GNAQ	100.0
GNAT1	100.0	GNB3	100.0	GPC3	100.0	GPC4	100.0
GPSM2	100.0	GRIA3	100.0	GRM6	100.0	GUCY1A1	100.0
GUCY2D	100.0	HEPACAM	100.0	HERC1	100.0	HLA-DQB1	100.0
HLA-DRB1	100.0	HRAS	100.0	HUWE1	100.0	IFT122	100.0
IFT140	100.0	IFT43	100.0	IGF1R	100.0	IHH	100.0
IL11RA	100.0	IRAG1	100.0	IRX5	100.0	JAG1	100.0
KAT6A	100.0	KCNH1	100.0	KCNV2	100.0	KDM1A	100.0
KDR	100.0	KIAA1109	100.0	KIF11	100.0	KIF7	100.0
KMT2E	100.0	KPTN	100.0	KRAS	100.0	L1CAM	100.0
LAMA1	100.0	LAMB1	100.0	LARGE1	100.0	LBR	100.0
LMX1B	100.0	LOC105377015	100.0	LRRIT3	100.0	LRP2	100.0
LRP5	100.0	LRPAP1	100.0	LRRC32	100.0	LTBP2	100.0
MASP1	100.0	MED12	100.0	MEGF8	100.0	MITF	100.0
MLC1	100.0	MMADHC	100.0	MPDZ	100.0	MSX2	100.0
MTM1	100.0	MTOR	100.0	MYCBP2	100.0	MYH11	100.0
NDE1	100.0	NF1	100.0	NFIA	100.0	NFIB	100.0
NFIX	100.0	NIN	100.0	NIPBL	100.0	NONO	100.0
NOTCH3	100.0	NPR2	100.0	NRAS	100.0	NSD1	100.0
ODCN	100.0	ODC1	100.0	OFD1	100.0	OPHN1	100.0
OSTM1	100.0	P4HB	100.0	PAFAH1B1	100.0	PDGFRB	100.0
PHF21A	100.0	PHF6	100.0	PIGA	100.0	PIGM	100.0
PIK3CA	100.0	PIK3R2	100.0	PKD1	100.0	POMGNT1	100.0
POMT1	100.0	POR	100.0	PPP1CB	100.0	PPP2R5D	100.0
PRDM5	100.0	PRIMPOL	100.0	PRPF6	100.0	PTCH1	100.0
PTEN	100.0	RAB23	100.0	RAB39B	100.0	RASA1	100.0
RECQL4	100.0	RELN	100.0	RIN2	100.0	RNF125	100.0
RNF135	100.0	RNF213	100.0	RUNX2	100.0	SAMHD1	100.0
SCARF2	100.0	SEC24D	100.0	SETD2	100.0	SH3PXD2B	100.0
SHANK3	98.74	SHOC2	100.0	SKI	100.0	SLC38A8	100.0
SLC39A5	100.0	SMAD3	100.0	SNX14	100.0	SOS1	100.0
SOX6	100.0	SPECC1L	100.0	SPRED1	100.0	STAMBP	100.0
STAT3	100.0	STRADA	100.0	SUFU	100.0	SUZ12	100.0
SYN1	100.0	SZT2	100.0	TBC1D7	100.0	TCF12	100.0
TCF20	100.0	TCOF1	100.0	TGFB2	100.0	TGFB3	100.0

TGFBF1	100.0	THSD1	100.0	TMCO1	100.0	TMEM94	100.0
TNFRSF21	100.0	TRAIP	100.0	TRIO	100.0	TRIP12	100.0
TRPM1	100.0	TSC1	100.0	TSC2	100.0	TSPAN12	100.0
TTC8	100.0	TUBA8	100.0	TUBB	100.0	TUBB2B	100.0
TWIST1	100.0	UPF3B	100.0	VCAN	100.0	VLDLR	100.0
VPS13B	100.0	VSX1	100.0	WDR19	100.0	WDR35	100.0
WDR62	100.0	WNT5A	99.83	YY1AP1	100.0	ZBTB20	100.0
ZBTB7A	100.0	ZEB2	100.0	ZIC1	100.0	MT-TF	100.0
MT-RNR1	100.0	MT-TV	100.0	MT-RNR2	100.0	MT-TL1	100.0
MT-ND1	100.0	MT-TI	100.0	MT-TQ	100.0	MT-TM	100.0
MT-ND2	100.0	MT-TW	100.0	MT-TA	100.0	MT-TN	100.0
MT-TC	100.0	MT-TY	100.0	MT-CO1	100.0	MT-TS1	100.0
MT-TD	100.0	MT-CO2	100.0	MT-TK	100.0	MT-ATP8	100.0
MT-ATP6	100.0	MT-CO3	100.0	MT-TG	100.0	MT-ND3	100.0
MT-TR	100.0	MT-ND4L	100.0	MT-ND4	100.0	MT-TH	100.0
MT-TS2	100.0	MT-TL2	100.0	MT-ND5	100.0	MT-ND6	100.0
MT-TE	100.0	MT-CYB	100.0	MT-TT	100.0	MT-TP	100.0